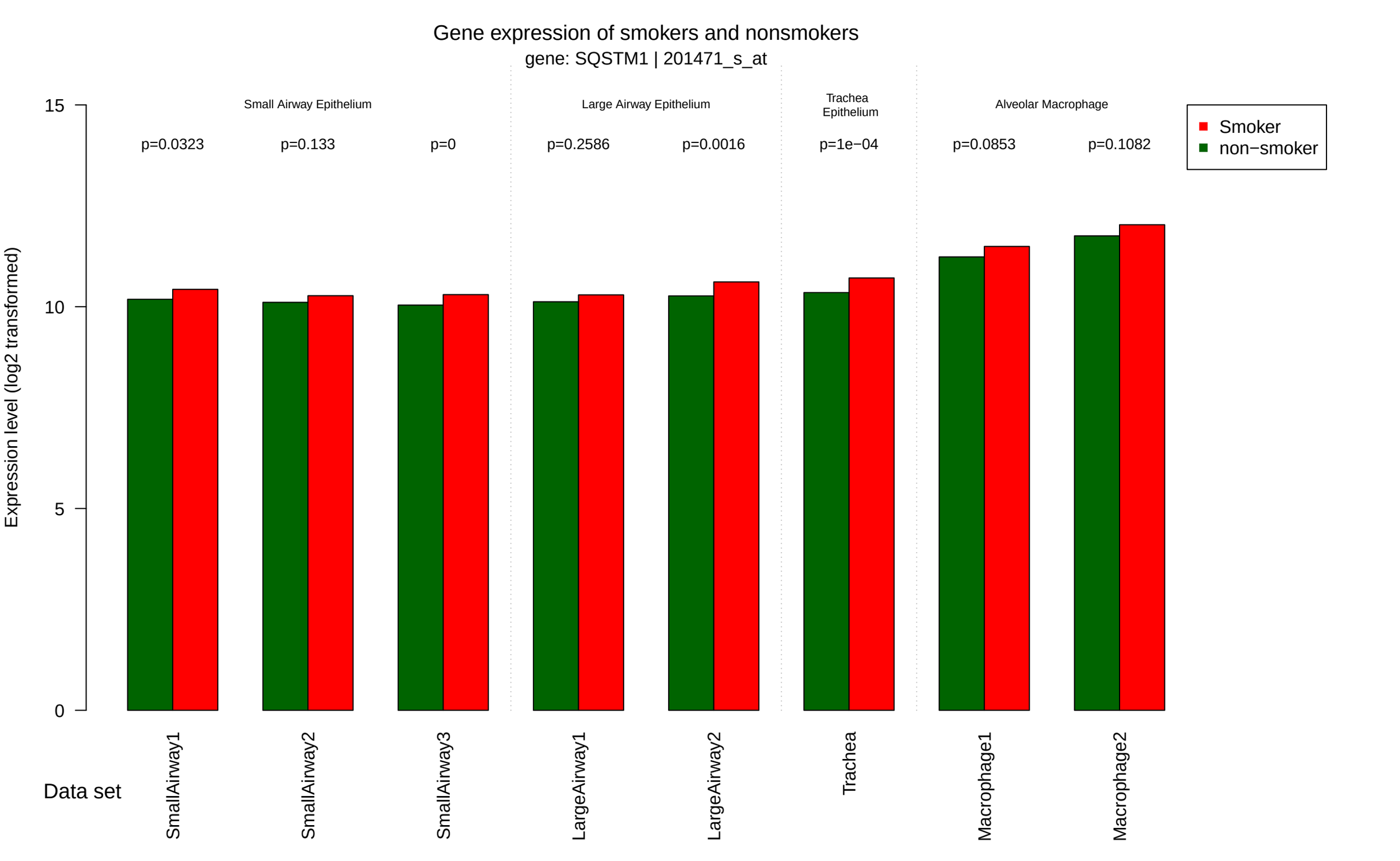
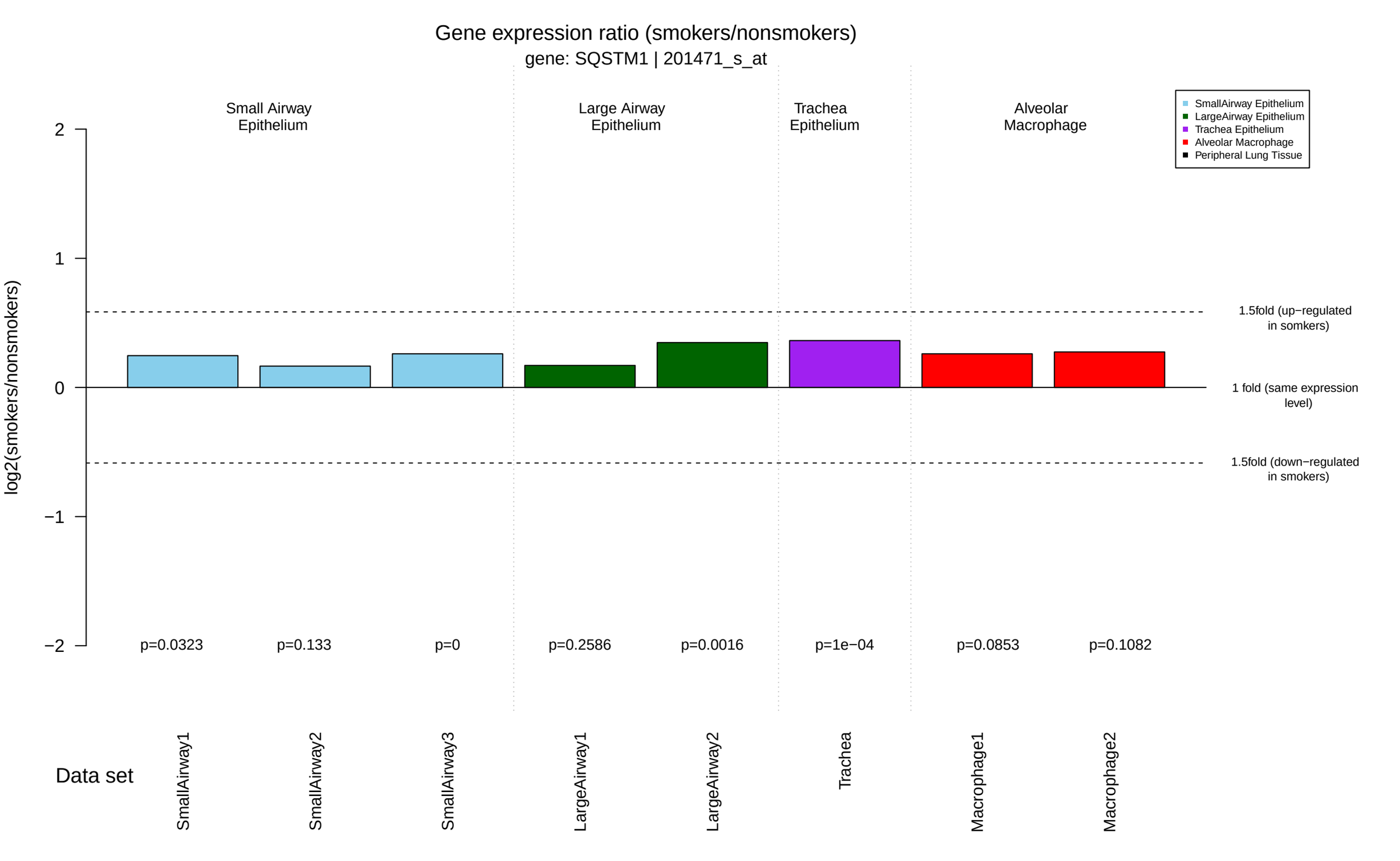
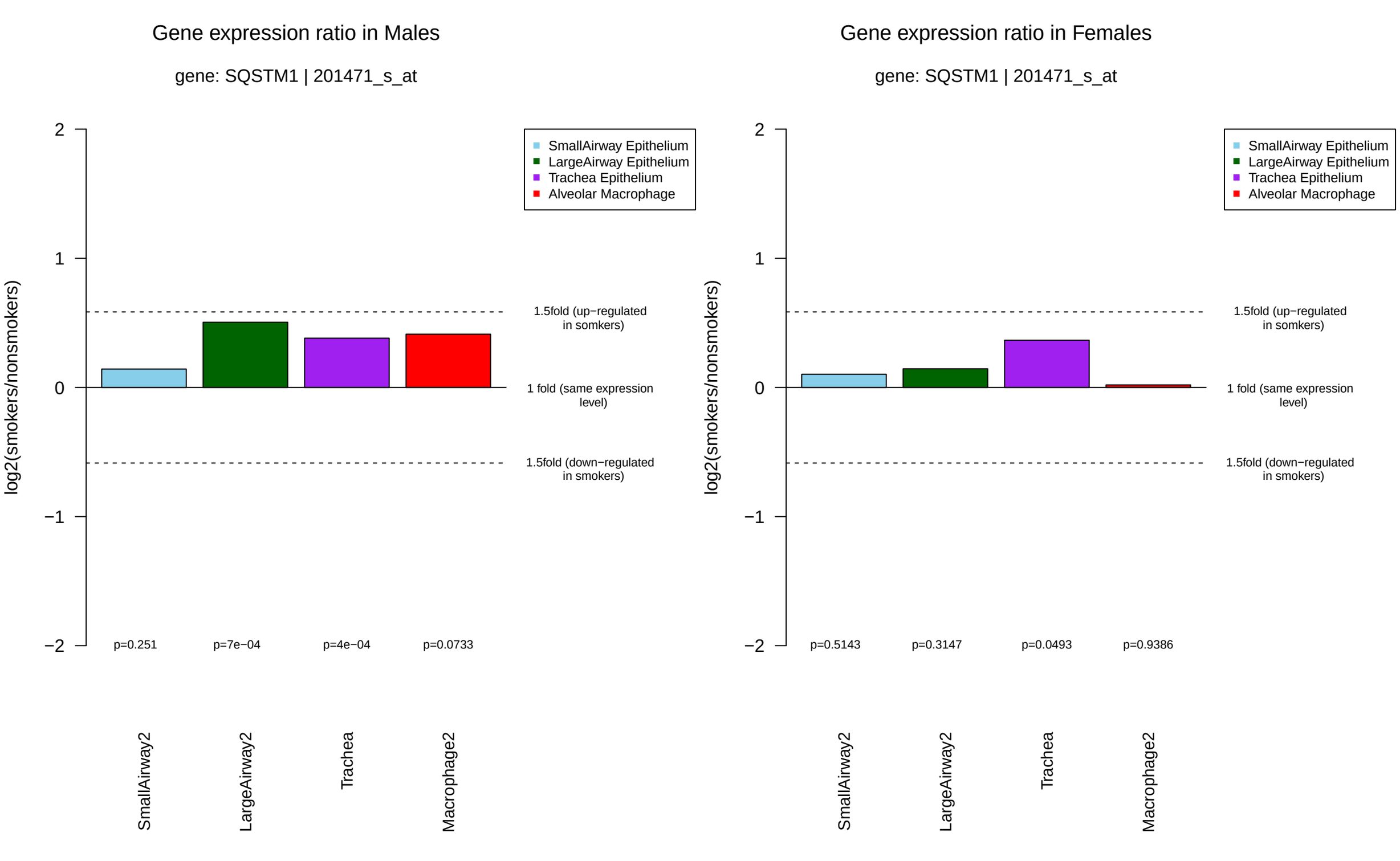
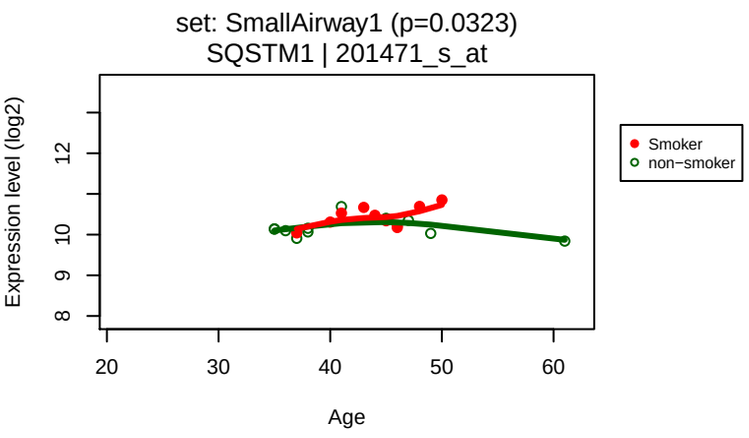
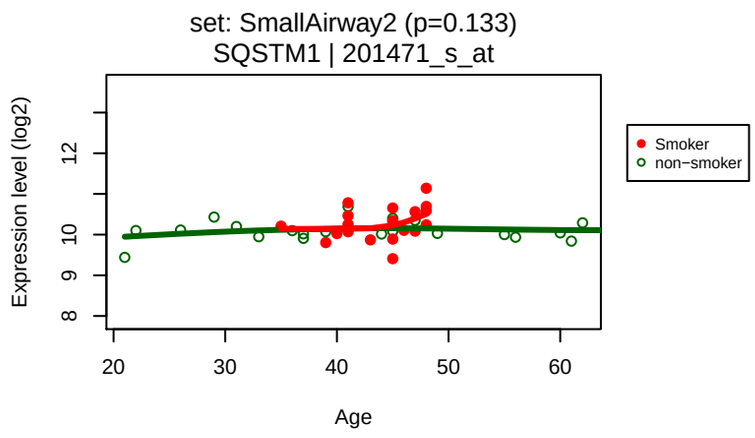
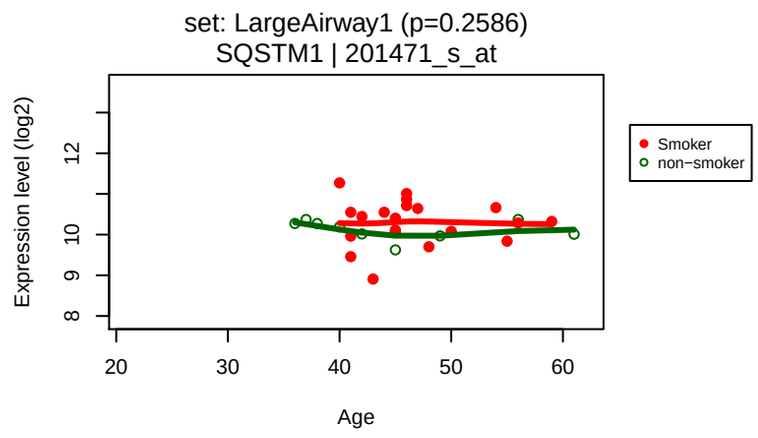

Gene expression of smokers and nonsmokers
gene: SQSTM1 | 201471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0323
p=0.133
p=0
p=0.2586
p=0.0016
p=1e−04
p=0.0853
p=0.1082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

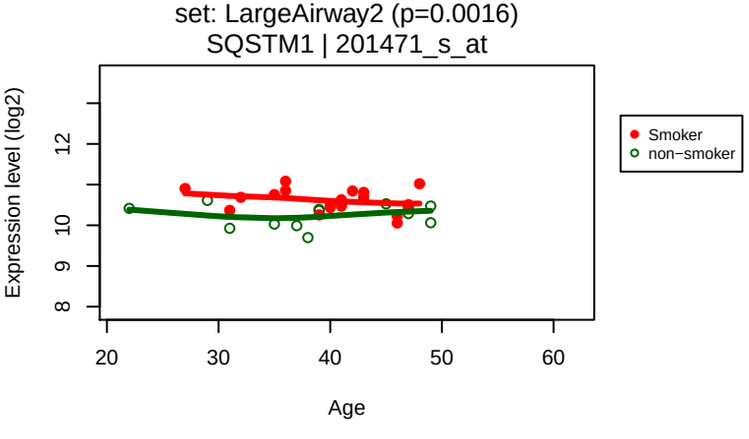

Gene expression ratio (smokers/nonsmokers)
gene: SQSTM1 | 201471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0323
p=0.133
p=0
p=0.2586
p=0.0016
p=1e−04
p=0.0853
p=0.1082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

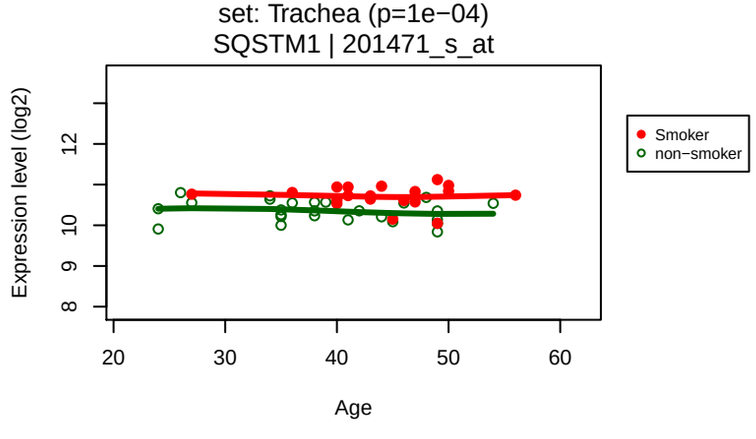

Gene expression ratio in Males
gene: SQSTM1 | 201471_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.251
p=7e−04
p=4e−04
p=0.0733
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SQSTM1 | 201471_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5143
p=0.3147
p=0.0493
p=0.9386
SmallAirway2
LargeAirway2
Trachea
Macrophage2

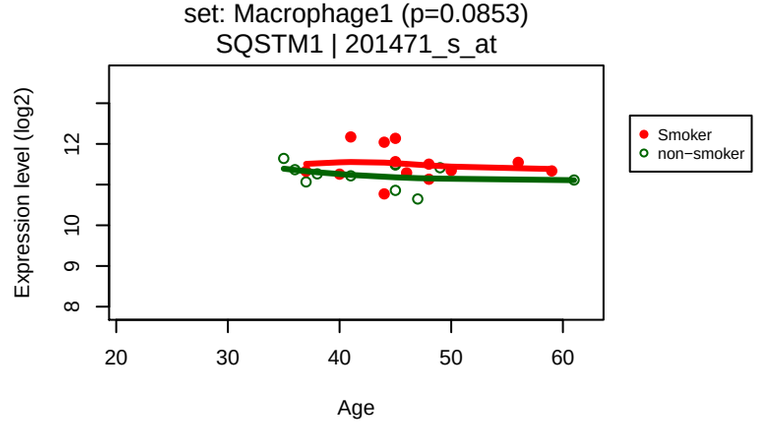

set: SmallAirway1 (p=0.0323)
SQSTM1 | 201471_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

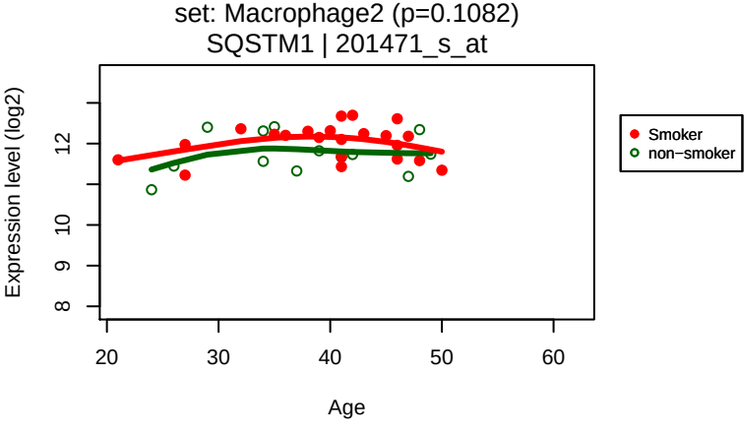

set: SmallAirway2 (p=0.133)
SQSTM1 | 201471_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

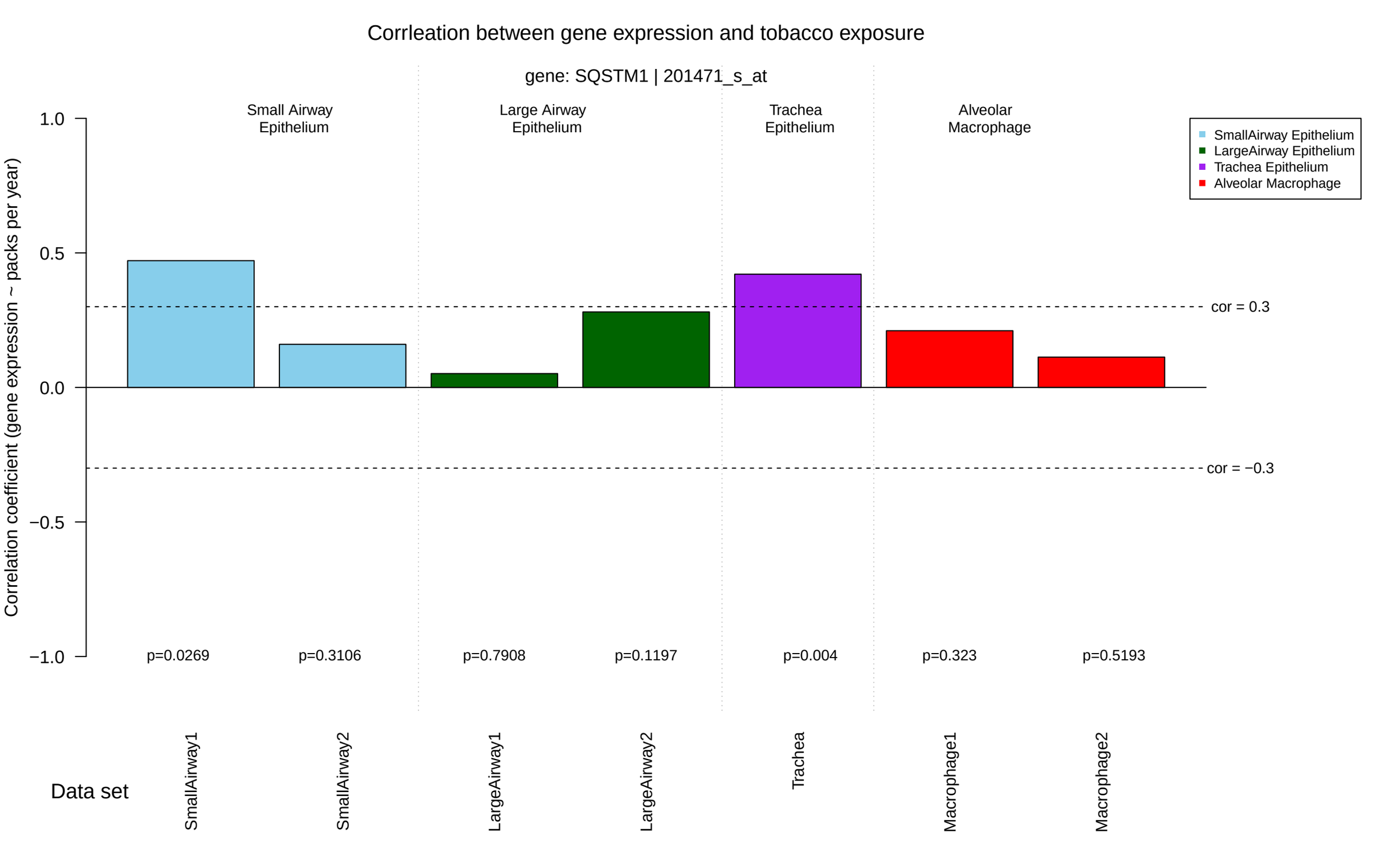

set: LargeAirway1 (p=0.2586)
SQSTM1 | 201471_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0016)
SQSTM1 | 201471_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
SQSTM1 | 201471_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0853)
SQSTM1 | 201471_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1082)
SQSTM1 | 201471_s_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SQSTM1 | 201471_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0269
p=0.3106
p=0.7908
p=0.1197
p=0.004
p=0.323
p=0.5193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2